

Investigating Natural Hydrogen Potential in the Bristol Channel–Bray Fault System via AI-Based Detection of Subcircular Depressions

Project Plan

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MSc Geo-Energy with Machine Learning and Data Science

<https://github.com/ese-ada-lovelace-2025/irp-wm722/>

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Project Summary

Discoveries of hydrogen (H_2) seeps worldwide have led to the recognition of natural H_2 as a potentially transformative energy source [1]. Hydrogen exploration is an emerging field, and recent discoveries of surface hydrogen leaks through 100 metre to kilometre-scale subcircular depressions (SCDs) [2, 3] have initiated an interest in AI-based remote sensing exploration methods. This MSc research project aims to develop and utilise deep learning methods to detect SCDs at an industrial exploration scale. Specifically, we will deploy exploratory neural networks to inform a field investigation in the UK Bristol-Bray fault corridor, where there is potential for natural hydrogen accumulations associated with Variscan faulting. The project will subsequently develop a comprehensive deep-learning workflow to scale up initial exploratory deep-learning techniques and field results to provide the first appraisal of natural hydrogen potential in the UK Bristol-Bray corridor.

Research Context

Hydrogen is expected to be a crucial resource in the energy transition, with 100 Mt already being produced globally in 2024, although less than 1% of this was low CO_2 -emissions H_2 [4]. Most hydrogen is currently produced by expensive and carbon-emitting processes such as steam-methane reforming (e.g., ‘grey hydrogen’). Natural hydrogen represents a promising carbon-free energy resource. Only 1×10^5 Mt of H_2 would provide more than double the amount of energy in all known natural gas reserves on Earth [5]. As such, accumulations of natural hydrogen in the UK would provide a hugely valuable domestic energy resource and would decrease reliance on carbon-intensive fossil fuels.

Exploration strategies for natural hydrogen are limited and rely on near-surface geochemical surveys and borehole data. However, subcircular depressions (SCDs) are promising indicators of subsurface H_2 migration and accumulation, and may be crucial for future resource exploration. The mechanism for SCD formation is poorly understood, but they have been proposed to form from mechanical compaction due to gas inflow and two-phase subsurface pore-pressure interactions [2]. Despite being a nascent field of research, recent studies [6, 7] have successfully employed pixel-wise segmentation deep learning approaches, utilising satellite imagery, to detect SCDs at scale. However, these approaches struggle to generalise across terrain types and land uses, as SCDs exhibit distinct optical surface expressions depending on climatic and agricultural conditions. Incorporating other remote sensing data causes problems with resolution and data availability, but may improve performance in certain geographies where optical techniques underperform. For example, recent studies investigating SCDs found that the structures can rise or subside at rates of up to 1 cm/yr [8], and can exhibit anomalies in multispectral indices [9].

Several SCDs have been found near the Bray Fault, a major Variscan strike-slip fault in the Paris Basin, France. Hydrogen concentration measurements along the Bray Fault have revealed variable but often considerably elevated values, reaching as high as 52 vol% in the Dogger Aquifer [10]. The Bray Fault forms part of the Variscan suture, which runs under the English Channel and connects with the Bristol Channel Fault Zone [11]. However, there is uncertainty regarding the continuity of H_2 generation and migration systems along the Bristol-Bray corridor, and the presence of a viable source rock.

Objectives

This project aims to investigate the promising potential of H₂ migration in the UK Bristol-Bray system. The research will employ deep learning, geostatistics, geochemistry, and structural geology to address the following core objectives:

- To complete exploratory deep-learning-based geostatistics with subsequent targeted fieldwork in the BCFZ and Lizard Complex, Cornwall, to evaluate tectonic structure and perform gas concentration measurement transects across pre-identified fault zones and potential SCD structures. Fieldwork will involve taking structural measurements to investigate fault connectivity and permeability to determine the migration potential of the fault zone.
- Develop a novel neural network, incorporating optical bands along with digital elevation data, topographic derivatives (e.g. multidirectional hillshade, topographic position index), and spectral indices (e.g. normalised difference vegetation index) to allow for efficient prediction at scale.
- Integrate field ground truth findings with deep learning techniques to refine detection performance, and evaluate model performance and results of deployment in southwest England.

Successful integration of the results of these objectives would allow for a confident initial appraisal of natural hydrogen resources in southwest England.

Methodology and Timeline

The project will be split up into three main sections (Figure 1):

1. The first three weeks of the project focus on fieldwork preparation, including a geological desk study and topography analysis to inform field targets.
2. Field work in the Bristol Channel Fault Zone (5 days) and Lizard Complex, Cornwall (2 days) from Saturday, June 13th.
3. Multi-channel convolutional neural network development, testing and deployment. This process will incorporate insights from field results to inform deployment targets and model design.

Calendar constraints have led to fieldwork being scheduled for the fourth week of the project. The first five days of the field work will take place near the Bristol Channel, where the tectonics show promise for hydrogen accumulations. The final two days will focus on the Lizard Complex, Cornwall, a Variscan ophiolite. As a result, instead of being used to confirm final model predictions, the field investigation will focus on the geological structure of the Bristol Channel region and performing soil gas measurements. Fieldwork will involve transecting major Variscan fault zones, including the Watchet-Cothelstone-Hatch Fault, to assess fault-related H₂ concentration variability, and conducting soil gas measurements at control locations and around suspected SCD structures. All soil gas measurements will be performed using two different techniques to account for drilling-induced H₂ artefacts observed in previous studies [12]. Although the field timing precludes the possibility of confirming the predictions of the final model, the field dates allow for the project to obtain important results at an early stage, which will inform

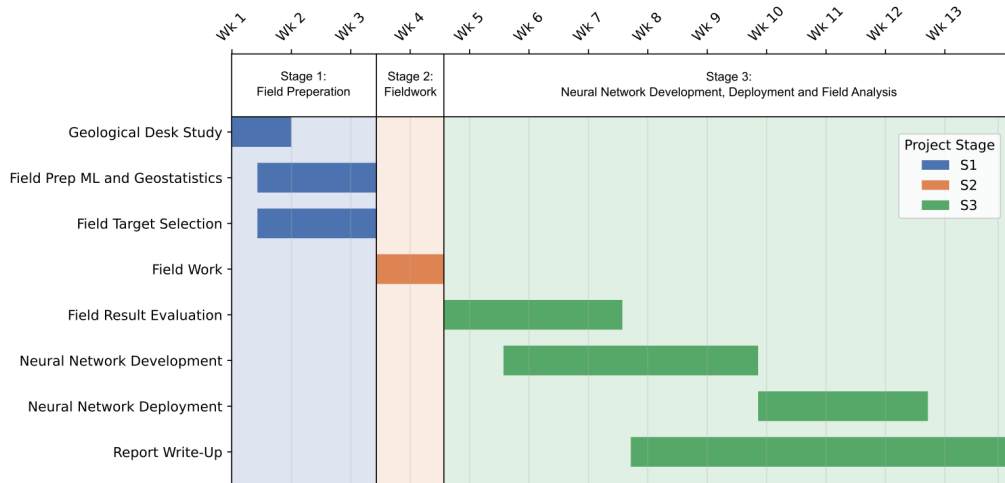


Figure 1: Gantt chart showing the three main planned stages of the project, and their major constituent parts.

the direction of the following work. Field insights will be incorporated into the development and deployment process to inform the appraisal of large target regions for SCDs. There is no available database for SCDs in the literature, so this must be slowly collated throughout the project. Convolutional neural networks are best suited for image-based problems in computer vision [13]. Several convolutional architectural types will be considered during the development of a neural network for this problem, including UNETs and Feature Pyramid Networks (FPNs).

Data Availability, Risks and Mitigations

The primary risks for this project relate to data availability and field work. Data availability and resolution constraints are the most pressing risks. While some countries have national-level open source Digital Elevation Models (DEMs) at 1 metre or 1/3 arc-second resolutions (e.g., UK, USA, France), many countries of interest lack national-level DEMs (e.g., Brazil, Namibia). Fortunately, ESA’s Sentinel-1 satellite provides DEM global coverage at 30 metre resolution. Although 30 metre resolution is too coarse for many relevant topographic applications, it does provide a valuable global dataset. Four-band spectral (R, G, B, NIR) data is available globally through Imperial’s Planet account [14]. Planet provides global coverage at 3 metre resolution, with a revisit time of 1-4 days. Initial modelling will utilise data at the globally available resolutions, and further experimentation with smaller but higher resolution datasets will inform the design of the final model. Soil gas measurements may not detect promising H_2 concentrations, indicating the lack of geologic hydrogen resources in the region. However, any results from the UK Bray Fault side are valuable in understanding migration systems in comparison to the H_2 -bearing Bray Fault in the Paris Basin. Field results will also provide direction for model deployment.

Early Work

Our early work, utilising deep learning to detect small depressions, shows that there is statistically relevant clustering of potential SCDs to Variscan NW-SE striking faults (Figures 2, 3). These results demonstrate the promising potential of H_2 accumulations in the region, and will form the basis of the targeted field work.

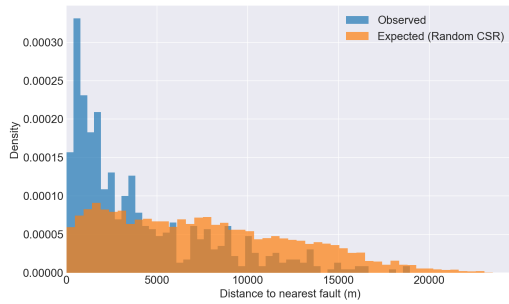


Figure 2: Distance-to-Fault histogram of topographic depression predictions within the investigated region.

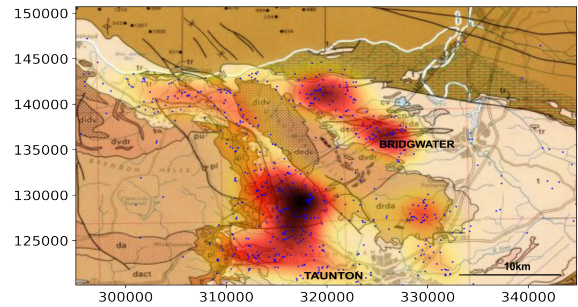


Figure 3: Gaussian kernel density estimation and scatter plot of depression predictions in the investigated region, with bedrock geology.

Outcomes

Successful completion of this project will result in the first investigation of natural hydrogen resources in the region. In particular, the project will form the first AI-informed evaluation of SCDs in the UK, and the first study directly investigating the potential presence of natural H_2 relating to Variscan structures in the UK. The successful incorporation of spectral and topographic derivative channels into a neural network will allow for the detection of SCDs across geographies and land uses. This would represent a meaningful improvement compared to current models, which are limited to particular climates and land uses, often being arid and non-agricultural regions.

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